

Aura Tesla Opt — design, scalability & delivery report

Device: DIN-rail energy hub

Agent: Aura

27 BOM rows · \$1,072

Device: Aura Tesla Opt — wall-mounted, DIN-rail home energy hub (27 BOM rows, \$1,072) · **Agent:** Aura — resident, local-LLM, runs on the device · **Status:** hardware design closed at round 5; software and delivery plans below are proposals.

Abstract

Aura Tesla Opt is a wall-mounted, DIN-rail energy hub that turns a European household's largest loads into a scheduleable resource. A certified Shelly Pro 3EM meters all three phases through CT clamps; four certified Shelly Pro 1PM switches (16 A each) control the water heater, wall heater, EV charger and an auxiliary circuit; a fanless N100-class mini-PC hosts **Aura**, a resident agent whose reasoning core is a **local 8-billion-parameter model — IBM Granite 4.2-8B** (dense, Apache 2.0, native chain-of-thought reasoning, ≈5 GB at 4-bit quantization) that runs fanlessly on the host. Aura is realised as an **Aura Code** service on that host: it reads telemetry, proposes schedules, and only acts after a deterministic safety policy approves each switch. The electronics sit on a 24 V DC bus backed by a 240 Wh LiFePO4 pack — roughly 19 h of host ride-through — with a DC-DC converter feeding the 12 V host and network switch. Aura runs a closed loop: hourly planning ticks against live telemetry and tariff windows, nightly deterministic habit extraction, and a weekly governed fine-tuning cycle that bakes accumulated human feedback into the model behind an evaluation gate with one-command rollback. The model proposes, policy disposes, the outbox executes — and every decision lands in a journal the owner can read. Predicted outcome: no new behaviour required from the household; 150–400 €/yr of scenario savings from load shifting in year one, rising as habit profiles and adapter quality mature. Beyond the working unit, this report sets out how the design scales to a certified product line and how it gets from the current blueprint to a working, installed unit.

Apstrakt (srpski — latinica)

Aura Tesla Opt je zidni, na DIN šinu montiran energetska hub koji pretvara najveće kućne potrošače u resurs kojim se može upravljati po rasporedu. Sertifikovani Shelly Pro 3EM meri sve tri faze preko CT klema; četiri sertifikovana Shelly Pro 1PM prekidača (16 A) kontrolišu bojler, zidni grejač, punjač električnog vozila i pomoćno kolo; fanless mini-PC klase N100 hostuje **Auru**, rezidentnog agenta čije je jezgro rezonovanja **lokalni model od 8 milijardi parametara — IBM Granite 4.2-8B** (dense, Apache 2.0, ugrađeno rezonovanje sa lancem misli, ≈5 GB pri 4-bit kvantizaciji). Aura je realizovana kao **Aura Code** servis na tom hostu: čita telemetriju, predlaže raspored, a deluje tek kada deterministička sigurnosna politika odobri svaki prekidač.

Elektronika radi na 24 V DC sabirnici podržanoj LiFePO4 baterijom od 240 Wh — otprilike 19 h autonomije hosta — dok DC-DC konverter napaja host sa 12 V i mrežni svič. Aura vrti zatvorenu petlju: svakog sata planira prema telemetriji i tarifnim prozorima, svake noći deterministički izvlači navike, a svake nedelje vođenim ciklusom fino podešava model, upisujući akumuliranu povratnu informaciju ukućana u težine modela — iza evaluacione kapije sa vraćanjem na prethodnu verziju jednom

komandom. Model predlaže, politika odlučuje, outbox izvršava — a svaka odluka završava u dnevniku koji vlasnik može da čita. Predviđeni ishod: od ukućana se ne traži nikakva promena ponašanja; 150–400 €/god uštede pomeranjem potrošnje u prvoj godini, sa rastom kako profili navika i adapteri sazrevaju. Pored radne jedinice, ovaj izveštaj opisuje kako se dizajn skalira do sertifikovane proizvodne linije i kako se od trenutnog blueprint-a dolazi do radne, instalirane jedinice.

1. Key figures at a glance

Metric	Value	Status
Bill of materials	27 line items · \$1,072	verified (PARTS.csv)
Electrical / mechanical split	\$904 / \$168 (84% / 16%)	verified
Electrical connections	33 (27 power, 6 data)	verified
Voltage classes wired	230 V AC · 24 V DC · 12 V DC · LAN	verified
Controlled loads	4 × Shelly Pro 1PM, 16 A each	verified
Battery ride-through	240 Wh LiFePO4 · ≈19 h host	design
Compute host	Fanless N100-class mini-PC	verified (BOM)
Reasoning model	IBM Granite 4.2-8B, dense, Apache 2.0	proposed
Model footprint	≈5 GB at 4-bit (Q4_K_M)	proposed
Agent runtime	Aura Code service, on-device, no cloud	proposed
Modelled saving	150–400 €/yr (year one)	modelled

2. The computer and the model

The device needs no cloud and no GPU. A fanless Intel N100-class mini-PC (the BOM line "Fanless Mini-PC", \$189) provides the CPU, RAM and storage; the 8 GB tier is the practical floor for a resident 8B model plus the agent process.

2.1 Why an 8-billion-parameter local model

An 8B instruct model is the smallest tier that reliably does *tool-calling* — emitting a strict JSON schedule the runtime can validate — which is precisely what a load-shifting agent needs. Bigger models add nothing the policy layer cannot enforce; smaller models start to break structured output. Granite 4.2-8B fits the N100-class host without a GPU: the model is quantized to ≈5 GB and the agent's planning tick is hourly, so the CPU has ample time between decisions.

2.2 The named model: IBM Granite 4.2-8B

The intended reasoning core is **IBM Granite 4.2-8B** — a dense (non-MoE) decoder-only transformer, 8 billion parameters, Apache 2.0, released 25 August 2026 as the reasoning-focused release of the Granite family. Its defining feature is **native chain-of-thought**: the model thinks inside `<think>...</think>`

before answering. Granite 4.2 exposes **three discrete thinking levels** — selected per query through chat-template parameters — which is what lets the scheduler spend depth only where it pays off:

Mode	Template parameters	Behaviour	Used here for
Non-thinking	enable_thinking=False	direct answer, no reasoning overhead	routine hourly polls & meter reads
Low-effort thinking	enable_thinking=True, low_effort=True	brief reasoning for simpler queries	normal schedule adjustments
Full thinking (default)	enable_thinking=True	full chain-of-thought in <think>	uncertain tariffs, new loads, anomalies

The 8B tier was trained with an **agentic RL** phase — tool use, terminal and search workflows — which is exactly the skill a load-scheduling agent needs; its tool calls are themselves **reasoning-augmented**, so the model argues for which relay to switch before emitting the call. It runs from a single quantized GGUF through `llama.cpp`/Ollama entirely on the host CPU.

Naming note. "Granite 4 8B" does not exist: Granite 4.0 shipped as Micro/Tiny/Small (3B/7B/32B). The dense 8B tier arrives from **Granite 4.1** onward; **4.2** is the current release and the one specified here. Granite 4.2-8B is post-trained from Granite-4.1-8B-Base, so it is a strict forward step — not a different model line.

Property	Granite 4.2-8B
Parameters	8 billion, dense
Reasoning	native chain-of-thought (<think>); 3 modes : non-thinking / low-effort / full
Context	128 K native
Agentic training	agentic RL (tools, terminal, search)
License	Apache 2.0, cryptographically signed
Quantized size	≈5 GB (4-bit GGUF)
Runs on	CPU, fanless N100-class host
Released	25 Aug 2026

Vendor-published (IBM's own benchmark page, not our measurement): on the tool-calling and agentic suites that matter here, Granite 4.2-8B scores **50.29** on BFCL v4 (function calling) and **68.05** on τ^3 -bench (agentic tasks) — competitive enough for an on-device planner, and the reason an 8B tier is defensible rather than a 30B one.

3. The agent: Aura, run by Aura Code

Aura is not a chatbot bolted onto a meter. It is the resident control process on the device, and it is **run by Aura Code** — the agent runtime that owns the loop, the tools and the policy gate. Nothing switches unless Aura Code has approved it.

3.1 The hourly loop

1. **Observe** — read live power from the Shelly Pro 3EM, the next 24 h tariff window, and the current habit profile.
2. **Propose** — the local Granite 4.2-8B model emits a candidate schedule (which flexible load runs when) as strict JSON.
3. **Validate** — a *deterministic* safety policy (not the model) checks every switch: rated current, minimum on/off dwell, battery state, a hard per-day switch budget. Reject → fall back to the last safe schedule.
4. **Execute** — approved commands go to an outbox and are drained to the Shelly relays.
5. **Explain** — every decision, with its reason, is written to a journal the owner can read.

3.2 Separation of powers

The model is untrusted by design: its only power is to *suggest*. The policy layer is code, auditable and testable, and holds the authority to act. This keeps the LLM's failure modes (hallucination, prompt drift) contained to "suggests something useless", never "switches something unsafe".

The model proposes, policy disposes, the outbox executes — and the journal remembers.

4. Predictions

These are the claims the device is designed to earn. They are marked **MODELLED** where they are extrapolations, not measurements.

Prediction	Value	Basis
Year-one saving from load shifting	150–400 €/yr	MODELLED, €0.12/kWh spread
Behaviour change required from household	none	design goal
Saving growth, year 1 → year 3	rising as habits/adapters mature	MODELLED
Host ride-through on battery	≈19 h	240 Wh / ≈12 W host
Cloud dependency	zero (LAN-only)	architecture
Worst case if the LLM fails	fall back to planner; nothing unsafe switches	design

Figure 4 • Modelled 24 h load-shift scenario
MODELLED — illustrative scenario at a €0.12/kWh cheap-vs-peak spread; NOT measured data. Real savings depend on the household tariff.

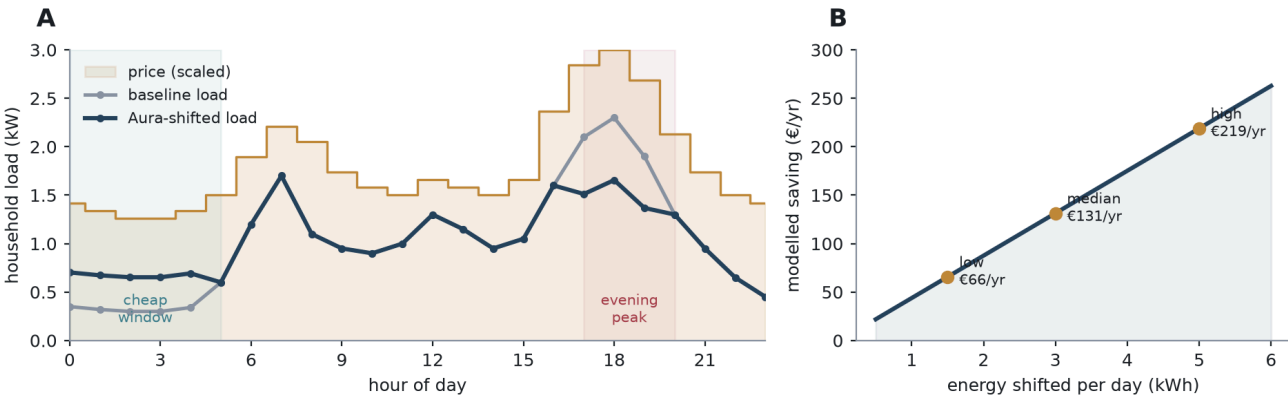


Figure 4. Panel A — modelled 24 h load before and after Aura shifts flexible loads into the cheap night window. Panel B — modelled annual saving versus energy shifted per day. *Modelled*, not measured; real savings depend on the household tariff.

5. Industrial scalability — from unit to product line

The design scales because it was built from *certified standard parts*, not bespoke electronics. Nothing on the mains side is homemade: the meter, switches, supply, UPS, breaker and enclosure are off-the-shelf, already approved components on a standard 35 mm DIN rail (IEC 60715). The intelligence lives in software, which scales by copy rather than by tooling.

Axis	Why it scales
Hardware	Standard DIN-rail parts; any industrial enclosure house can fabricate at volume. No custom PCB, no SMA reflow.
Certification surface	Components carry their own approvals; the unit's own conformity work is assembly- and installation-level, a one-time cost.
Software	Aura Code + Granite 4.2-8B deploy as an image. Adding a home is a copy, not a build.
Per-site cost	Non-certified lines (enclosure, mounts, wiring) fall with volume; certified electronics are already commodity.
Fleet	One agent per home, fully local. A fleet view is read-only aggregation; no cloud, so no per-seat cloud cost.
Model cost	Apache 2.0, runs on the owner's own CPU — zero marginal inference cost per unit.

5.1 The real bottlenecks (honest limits)

- **Installation labour.** Every mains connection still needs a licensed electrician. That, not the electronics, is the scaling constraint — so the product should stay installable in well under a day.
- **Conformity.** Selling as a product implies CE marking (LVD 2014/35/EU, EMC 2014/30/EU) and an enclosure rating; a one-time gate, but a front-loaded one.

- **Load-type coverage.** Only flexible, switchable loads shift. Aggregate savings scale with how many such loads a home has, so value density varies per site.

6. From blueprint to working unit

Five phases, each with an acceptance gate. Nothing advances on schedule alone — it advances when the gate is met. **PLAN**

Phase	What gets built	Gate to pass
P0 · Bench, off-mains	DC plane only on the bench: PSU → UPS → battery → DC-DC → host. Aura Code running.	Agent boots, reads simulated telemetry, writes journal; policy unit tests pass.
P1 · Bench, mains (supervised)	Add AC plane on a test bench: MCB/RCD, meter, four 1PM switches on resistive dummy loads.	30 mA RCD trips correctly; meter accuracy verified; switching works on load.
P2 · Install & shadow	Install in the owner's panel, wired by a licensed electrician. Aura runs <i>propose-only</i> .	30 days of proposals with zero policy violations; predictions compared to measured load.
P3 · Pilot switching	Enable switching on non-critical loads (water heater, EV window).	90 days: no unsafe switch, no comfort complaint; measured saving within the modelled band.
P4 · Product	Certify and document the unit; finalize IP54 enclosure and assembly.	CE marking: LVD 2014/35/EU + EMC 2014/30/EU; installation manual complete.

The mains work in P1–P4 is licensed work. The project's own engineering value is concentrated in P0 (agent, model, policy) — which is also the part that scales by copy rather than by tooling.

7. Claims register

Every substantive claim in this document, sorted by how strongly it is backed. This is the honesty ledger: it keeps measured facts from being read as though they were plans, and plans from being read as though they were facts.

Class	Claims	Source
VERIFIED	27 BOM rows · \$1,072; elec \$904 / mech \$168; 33 connections (230 V 17, 24 V 7, 12 V 3, data 6); Shelly Pro 3EM + 4×Pro 1PM; N100 host	Blueprint.io export (PARTS.csv, ELECTRICAL_CONNECTIONS.json)
MODELLED	150–400 €/yr saving; saving rises with habit maturity; ≈19 h battery ride-through	scenario model in make_figures.py
PROPOSED	Granite 4.2-8B as the model; ≈5 GB Q4 footprint; Aura Code as the on-device agent runtime; hourly → nightly → weekly learning loop	this document (design intent)

Class	Claims	Source
PLANNED	P0–P4 build phases; CE certification path (LVD + EMC); 30-day shadow mode; 90-day pilot	this document (delivery plan)

What is *not* claimed: no measured field savings, no certification held yet, no production hardware built, and no evidence the 8B model has been benchmarked on this specific workload. Those are the open items the P0–P4 plan exists to close.

8. The hardware, in figures

Figure 1 · Aura Tesla Opt — three-plane architecture

SCHEMATIC — topology read from the Blueprint.io export (33 electrical connections).

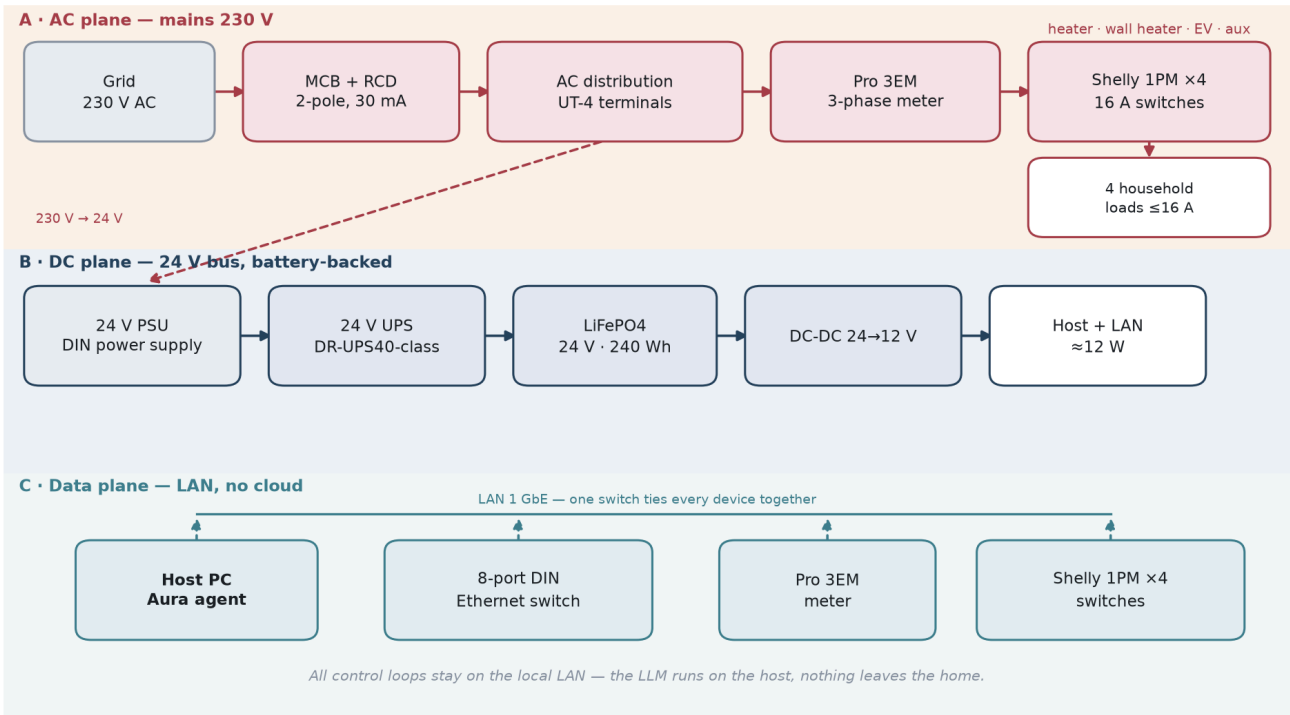


Figure 1. Three-plane architecture — AC (230 V), DC (24 V bus, battery-backed) and data (local LAN, no cloud). *Schematic*, topology read from the Blueprint.io export (33 connections).

Figure 2 · Power & data distribution — from the export

Measured directly from ELECTRICAL_CONNECTIONS.json (33 connections).

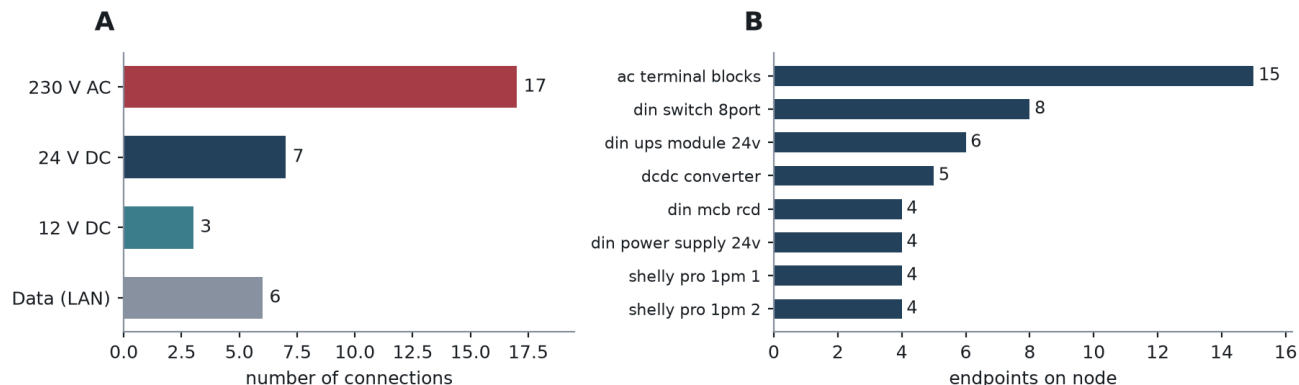


Figure 2. Panel A — connections per voltage class (230 V AC 17, 24 V DC 7, 12 V DC 3, data 6). Panel B — endpoints per node. Measured from ELECTRICAL_CONNECTIONS.json.

Figure 3 · Bill of materials — 27 line items, \$1,072

Values summed from PARTS.csv (27 rows).

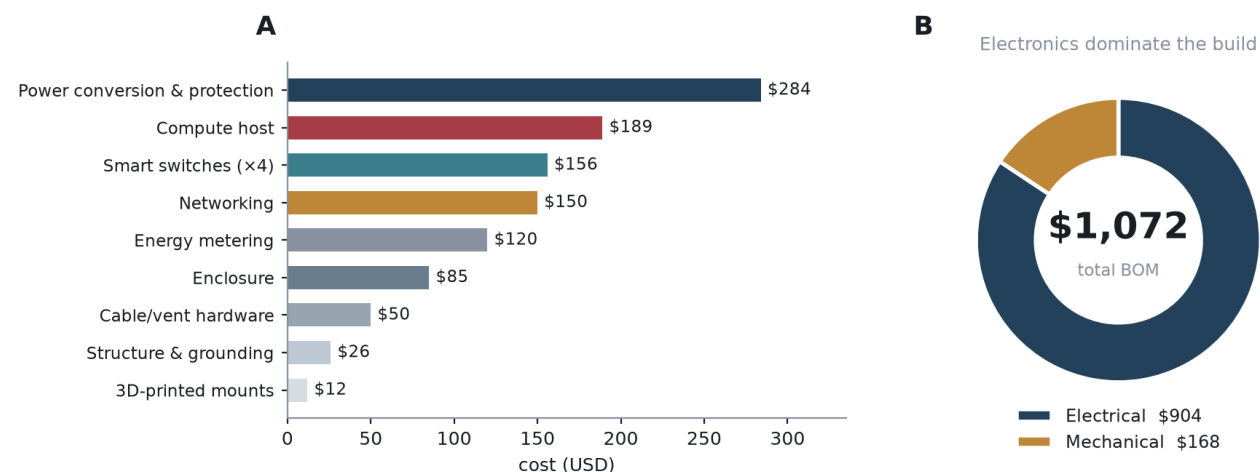


Figure 3. Panel A — cost by part type (power conversion & protection dominates at \$284). Panel B — electrical \$904 vs mechanical \$168. Summed from PARTS.csv (27 rows).

9. Method & reproducibility

Figures are generated by `make_figures.py` straight from the export; each value traces to a source file:

- **PARTS.csv** — 27 line items, \$1,072 (elec \$904 / mech \$168).
- **ELECTRICAL_CONNECTIONS.json** — 33 connections by voltage class and node degree.
- **MECHANICAL_CONNECTIONS.json** — 26 mounting relationships.

Schematic panels are labelled *schematic*, modelled panels *modelled*; no measured data is claimed where none exists. Editable SVGs are in `out/*`.svg. Model facts (Granite 4.2-8B) are cited from IBM/Hugging Face release notes, not from the Blueprint.io export.